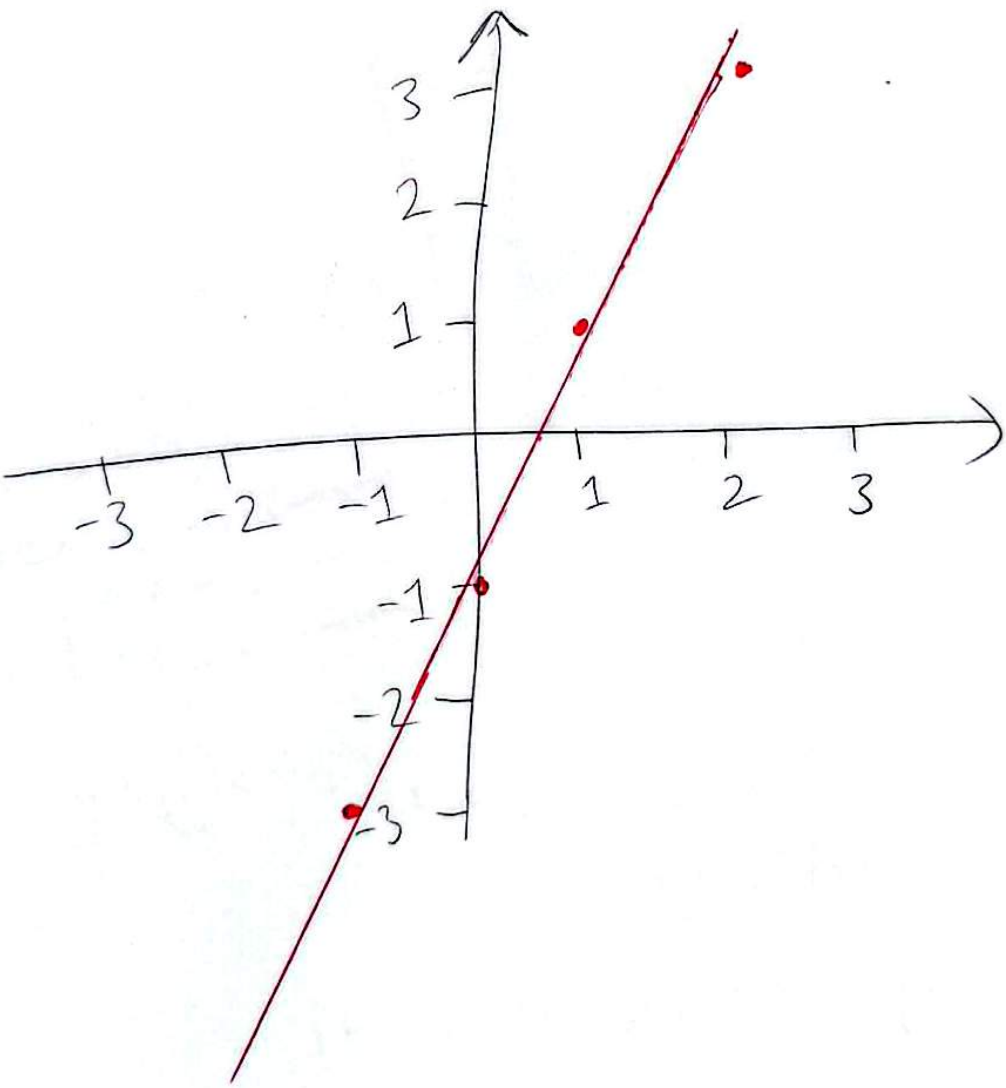
Graphs of Equations

e.g. sketch graph of $y=2x-1$

x	-1	0	1	2
y	-3	-1	1	3



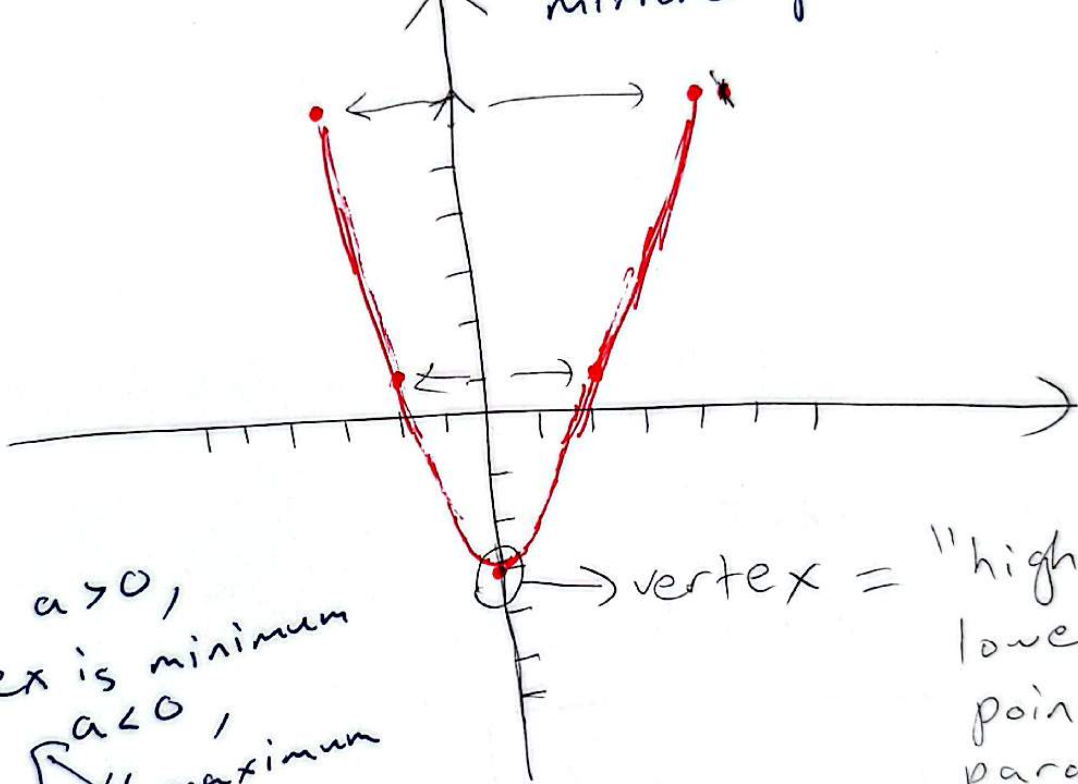
g. Sketch $y=x^2-3$

mirrored pair

vertex point

x	-3	-2	0	2	3
y	6	1	-3	1	6

mirrored pair



vertex = "highest or lowest point" on parabola

when $a>0$, vertex is minimum
" $a<0$, " maximum

$y = ax^2+bx+c$ "bowl-shaped" specifically parabola

if a is positive, parabola "opens up"

" a " negative, " "opens down"

J. Find the x-intercepts & y-intercepts of $y=x^2-3$

To solve for x-int.s, set $y=0$

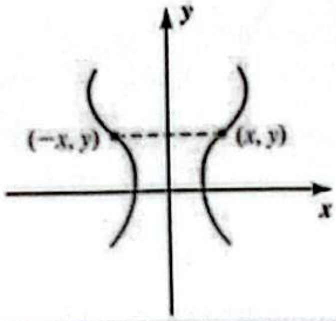
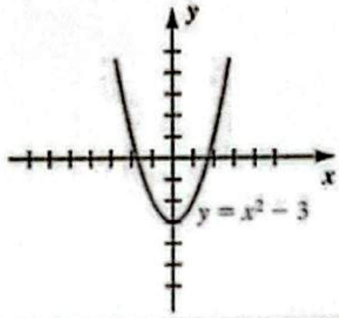
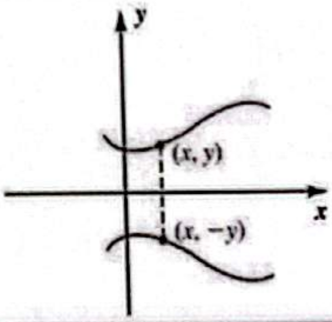
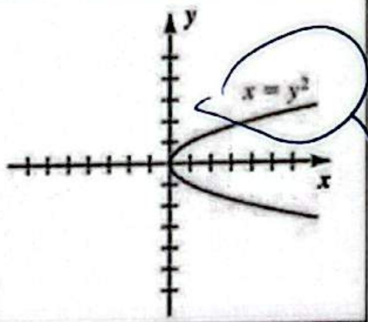
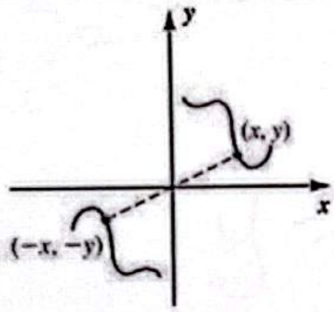
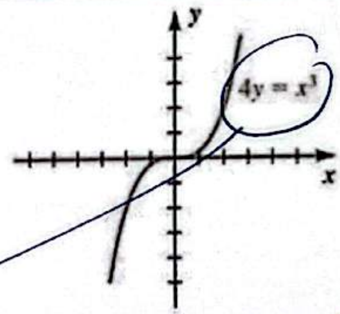
To solve for y-int.s, set $x=0$

$$y=0 \iff x^2-3=0$$
$$\iff \sqrt{x^2}=\sqrt{3}$$
$$\iff x=\pm\sqrt{3}$$

$\boxed{(\sqrt{3},0) \text{ and } (-\sqrt{3},0)}$

$$x=0 \iff y=-3$$

$\boxed{(0,-3)}$

Terminology	Graphical interpretation	Test for symmetry	Illustration
The graph is symmetric with respect to the y-axis.		**(1)** Substitution of $-x$ for x leads to the same equation.	
The graph is symmetric with respect to the x-axis.		**(2)** Substitution of $-y$ for y leads to the same equation.	
The graph is symmetric with respect to the origin.		**(3)** Simultaneous substitution of $-x$ for x and $-y$ for y leads to the same equation.	

$$4y = x^3$$

$$\updownarrow$$

$$4(-y) = (-x)^3$$

$$-4y = -x^3$$

$$4y = x^3$$

$$x = f(y)$$

$$x = y^2$$

replace y by $-y$

$$x = (-y)^2$$

$$x = y^2$$

$y^2 = x$ $\qquad x = f(y)$

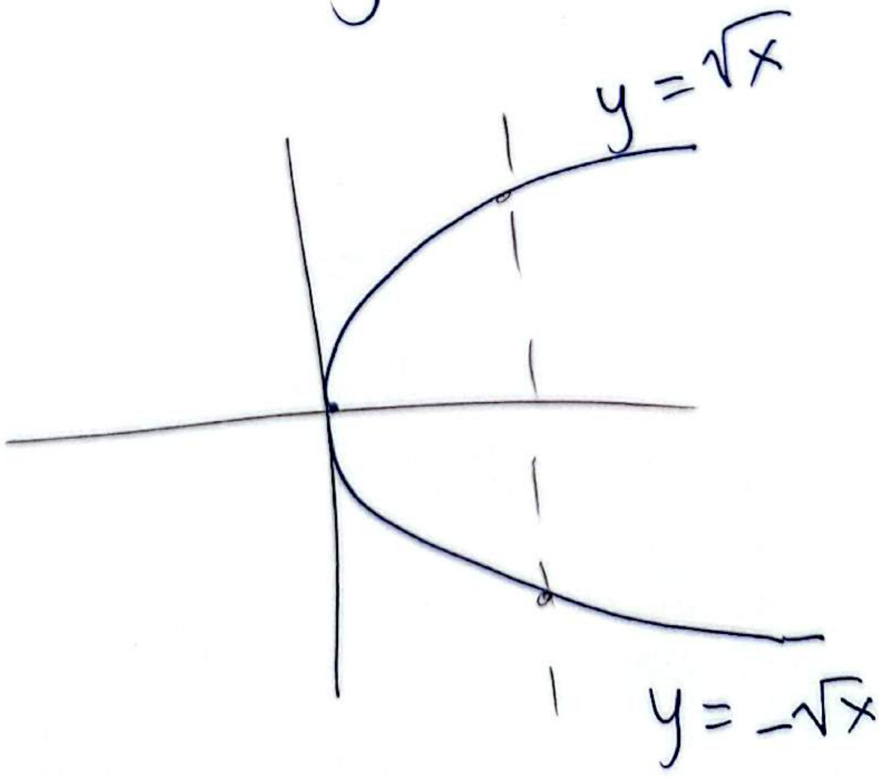
x	4	1	0	1	4
y	-2	-1	0	1	2

vertex

$y = f(x)$?

$y^2 = x \longrightarrow y = f(x)$

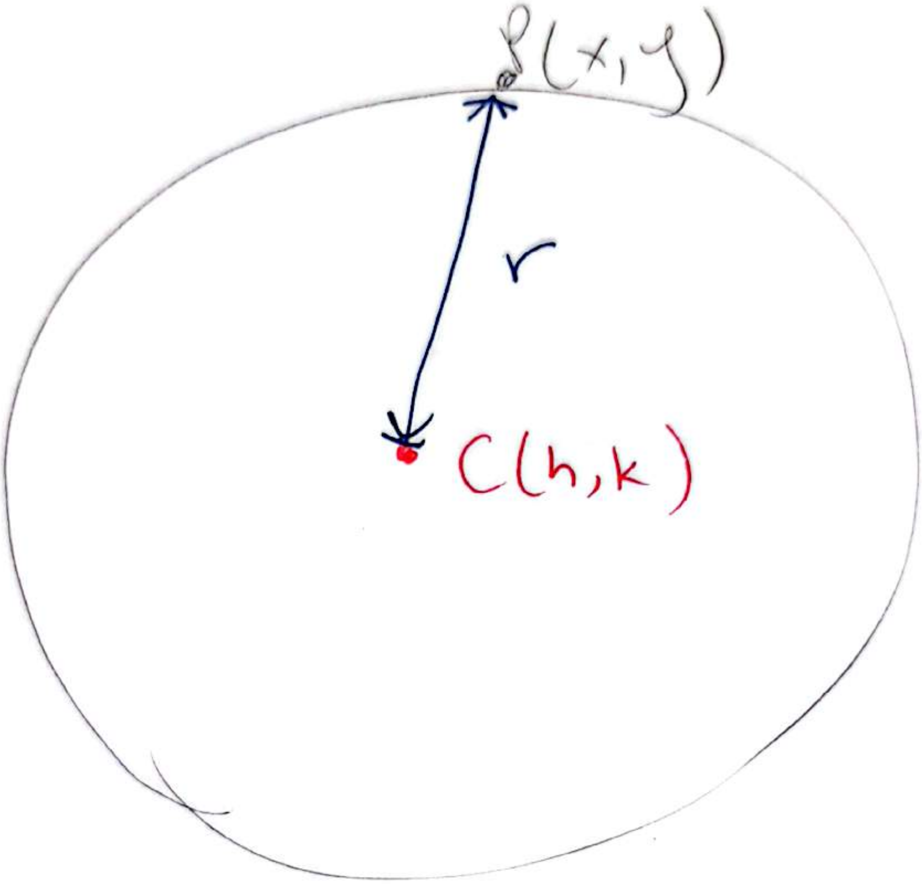
$y = \pm\sqrt{x}$



5. $4y = x^3$ $\qquad$ $y = \frac{x^3}{4}$

x	-2	-1	0	1	2
y	-2	$-\frac{1}{4}$	0	$\frac{1}{4}$	2

Equation of Circle



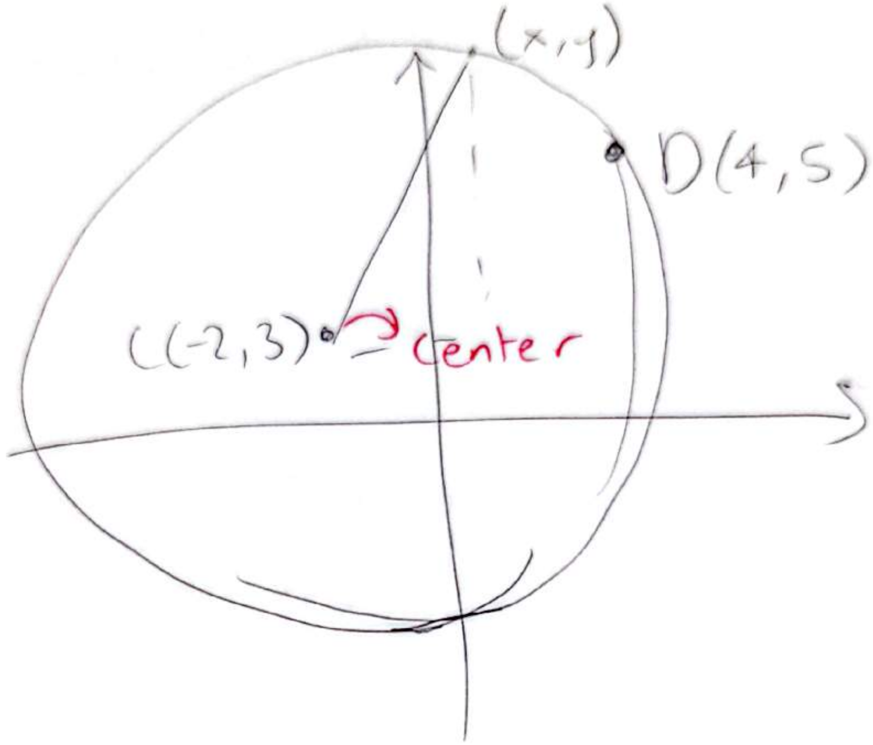
$$(x-h)^2+(y-k)^2=r^2$$

$$\Updownarrow$$

$$\underbrace{\sqrt{(x-h)^2+(y-k)^2}}_{\downarrow}=r$$

$$D(P,C)=r$$

8. Find the equation of the circle sketched below



$$r = d(C, D)$$

$$= \sqrt{(4-(-2))^2 + (5-3)^2}$$

$$= \sqrt{6^2 + 2^2} = \sqrt{40}$$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-(-2))^2 + (y-3)^2 = (\cancel{\sqrt{}}40)^2$$

$$\boxed{(x+2)^2 + (y-3)^2 = 40}$$

9. Find the center and radius of the circle with equation

$$3x^2+3y^2-12x+18y=9$$

$$\Big\downarrow \div 3$$

$$x^2+y^2-4x+6y=3$$

$$(x^2\underbrace{-4}_{\div 2\,=\,-2}x)+(y^2\underbrace{+6}_{\div 2\,=\,+3}y)=3$$

$$(x-2)^2+(y+3)^2+d=3$$

$$(x^2-4x+4)+(y^2+6y+9)+d=3$$

$$d=-4-9=-13$$

$$(x-2)^2+(y+3)^2-13=3$$

~~center(-2,3)~~

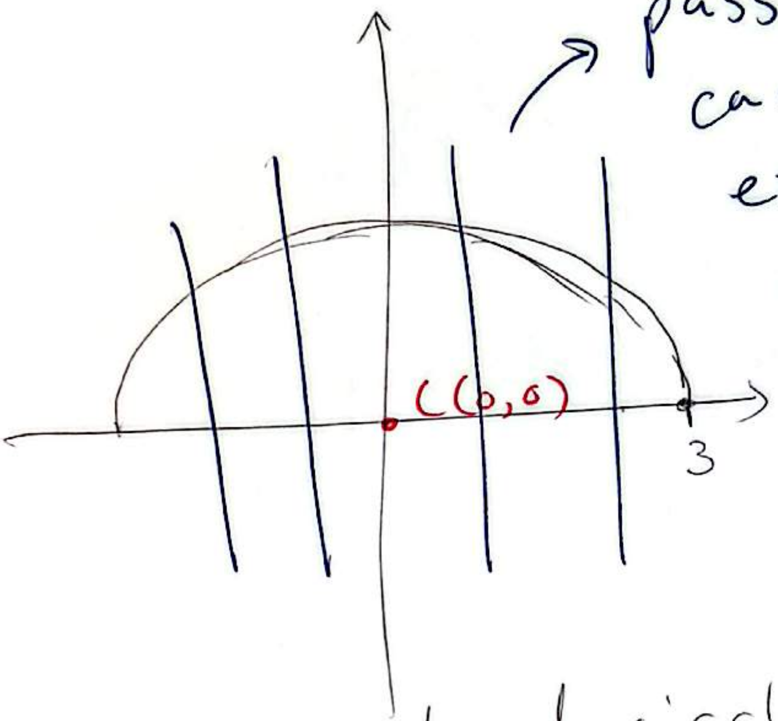
center(2,-3)

$$(x-2)^2+(y+3)^2=16$$

radius $=4$

$$(x-2)^2+(y-(-3))^2=4^2$$

9. Find an equation for the semicircle:



passes VLT can be expressed as $y = f(x)$

Start w/ standard circle equation

$$(x-0)^2 + (y-0)^2 = 3^2$$

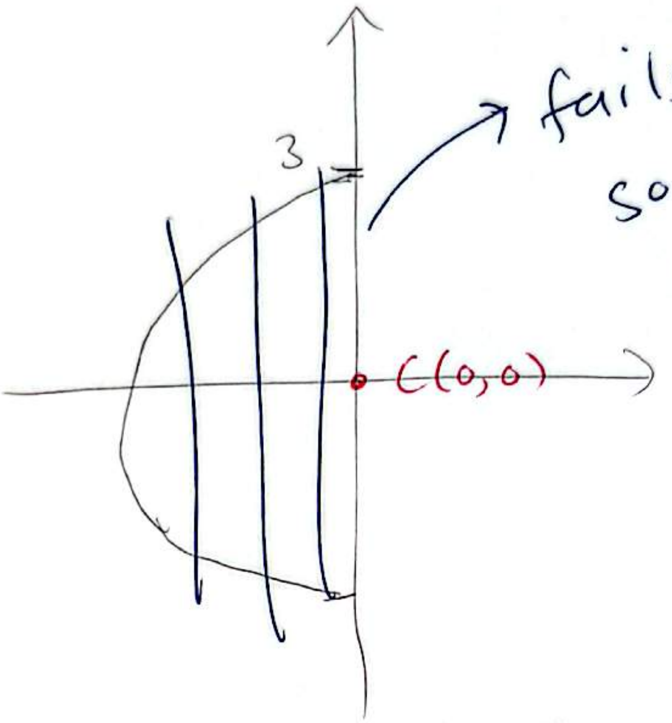
$$x^2 + y^2 = 9$$

$$y^2 = 9 - x^2$$

$$\boxed{y = \sqrt{9 - x^2}}$$

If you say $y = \pm\sqrt{9-x^2}$ this is equation of full circle

Find an equation for the semicircle:



fails VLT so we can't do $y=f(x)$ but we can $x=f(y)$

$$(x-0)^2+(y-0)^2=3^2$$

$$x^2+y^2=9$$

$$x^2=9-y^2$$

$$\boxed{x=-\sqrt{9-y^2}}$$

If we wrote $x=+\sqrt{9-y^2}$ we would the right-half of the circle instead.